

Name: I Forgot Team Date: _____

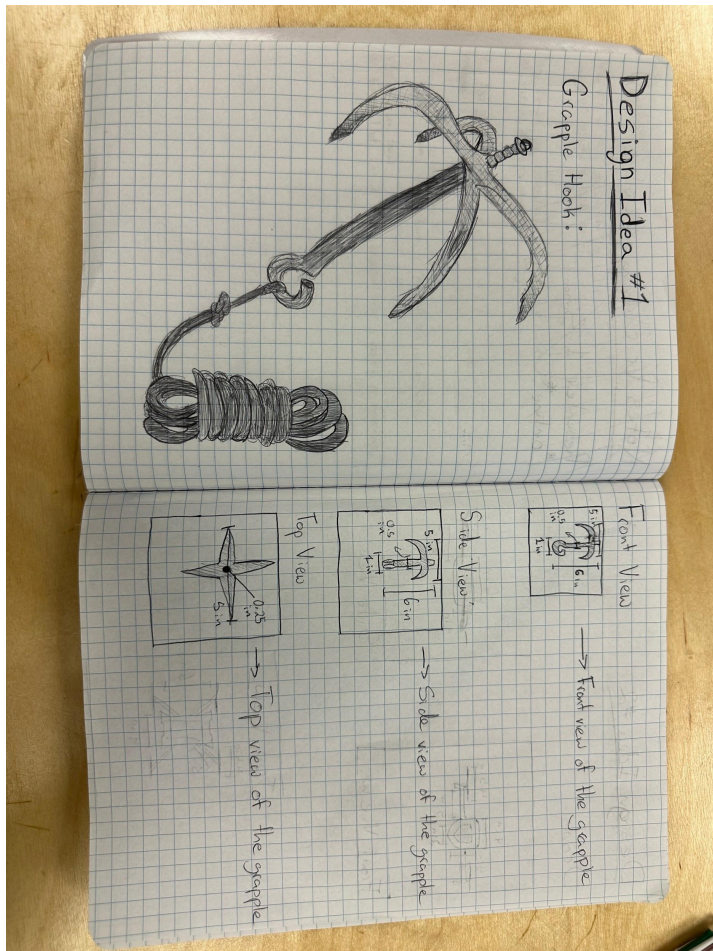
Class: _____

MyDesign Portfolio

Element D: Design Concept Generation, Analysis, and Selection

Design Idea Generation (D1)

Grapple Hook



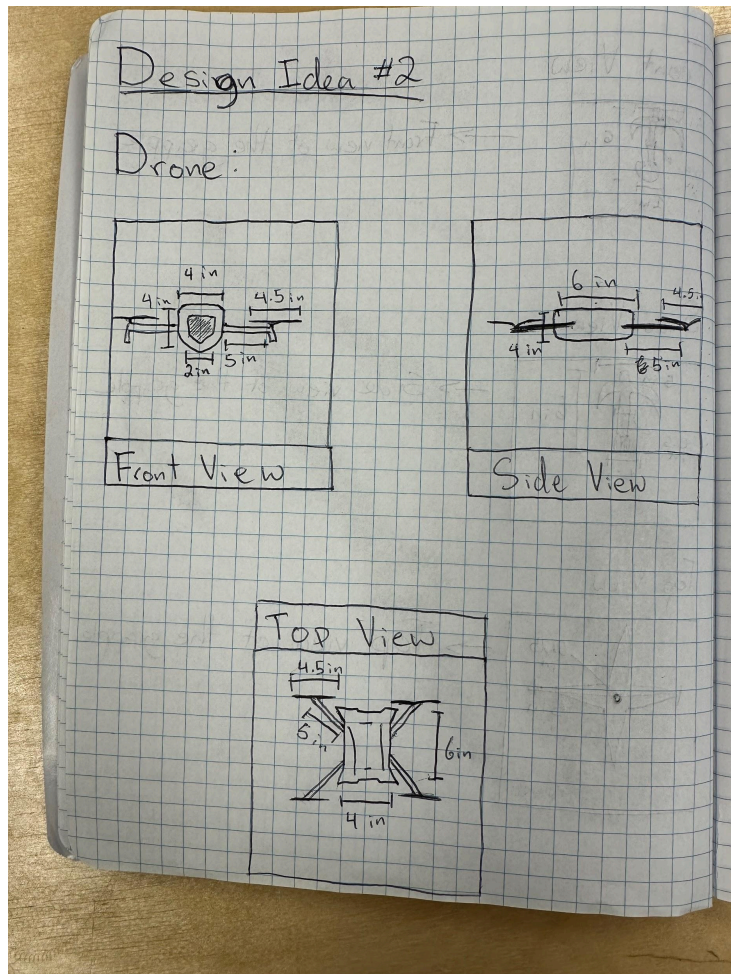
Date: 2/1/24

Description: This tool features a metal weighted hook at the end of a proper-length rope. Once thrown by the user, it will latch onto any badminton birdies on the surface and will be

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pulled by the grapple so it falls down.

Drone



Date: 2/5/24

Description: This device will be remotely controlled and will fly up to the badminton birdies. The birdies will be pushed by the drone and will fall down and the drone could fry from place to place.

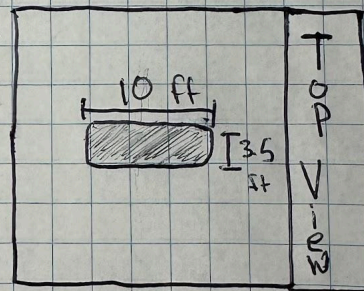
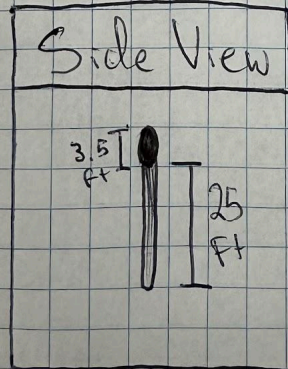
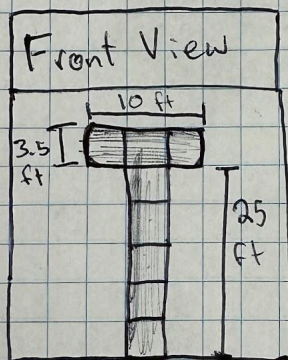
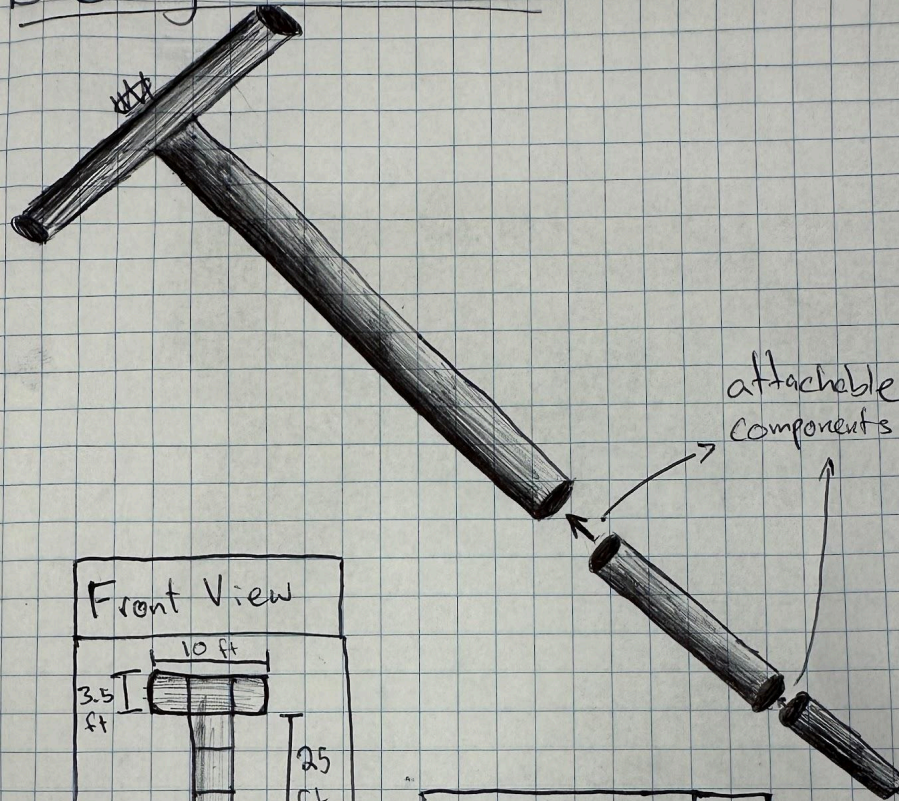
Drawing that includes distinguishing information such as scale, dimensions, and materials:

Long Stick

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Design Idea #3



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Date: **2/5/24**

Description: This tool will be the proper length of the gym and will be long enough to reach the birdies. Once lifted up, the stick will be able to poke all of the badminton birdies down from the windows and raftors.

Drawing that includes distinguishing information such as scale, dimensions, and materials:

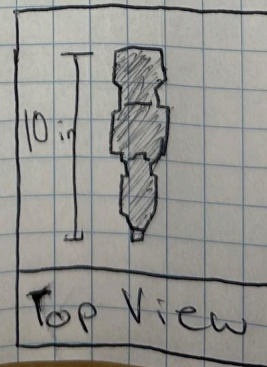
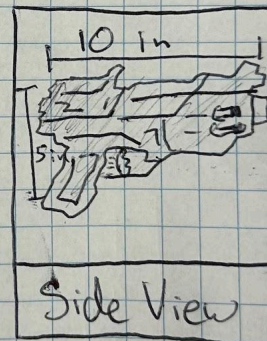
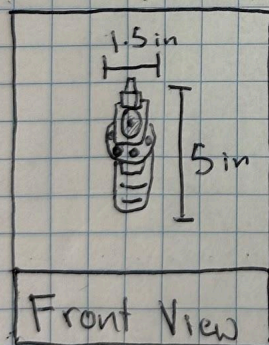
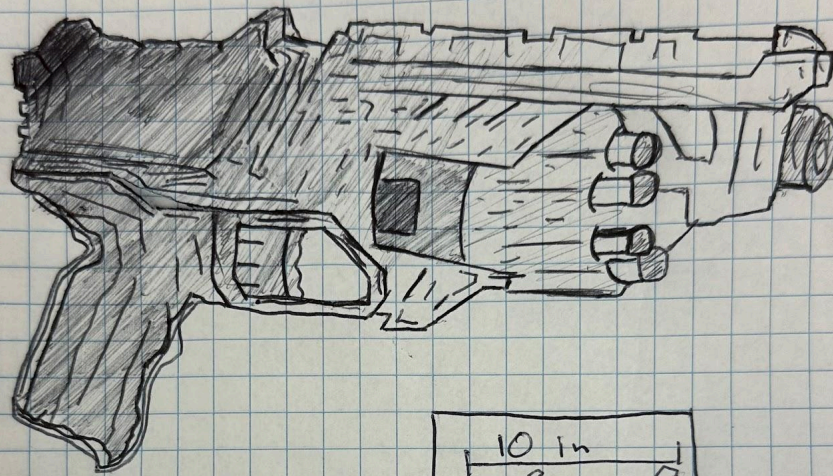
Nerf Gun

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Design Idea #4

Nerf Gun:



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Description: Although it is a toy, it can be used to shoot down any birdie from its position with aiming and shooting the foam darts at the birdies. The birdies will fall down after contact from the darts and will get the birdies down.

Drawing that includes distinguishing information such as scale, dimensions, and materials:

Design Idea Comparison and Selection (D1, D2)

	Weight	Grapple Hook	Drone	Stick	Nerf Gun
Height - Can it reach the birdies?	10	5	10	10	5
Transportation - can it be transported?	5	5	5	5	5
Functionality - does it work?	15	10	10	15	0
Stability - is it stable/durable?	5	5	5	5	0
Efficiency - can it be done quickly?	10	5	5	10	0
score	45	30	35	45	10

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Design Blueprint (D4)

Describe all of your design's details such that someone unfamiliar with your project could build your prototype. Include sketches, dimensions, CAD (if taught in your class), manufacturing methods, material choices, supply lists, and costs.

Design Blueprint #1

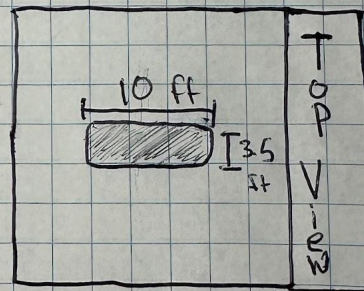
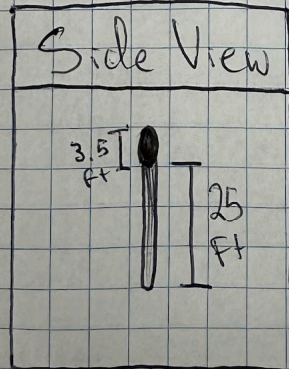
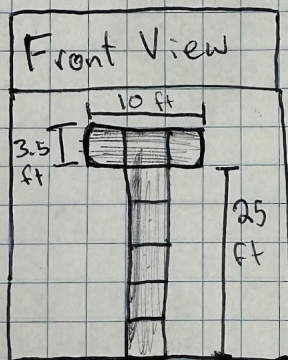
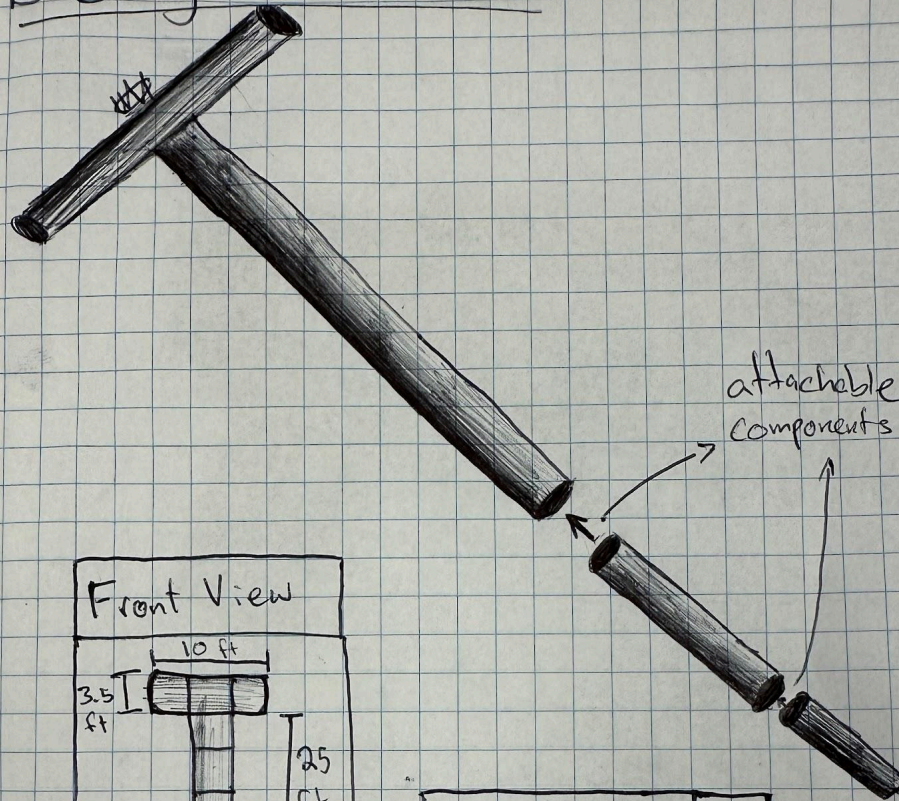
Date: **2/5/24**

Sketch:

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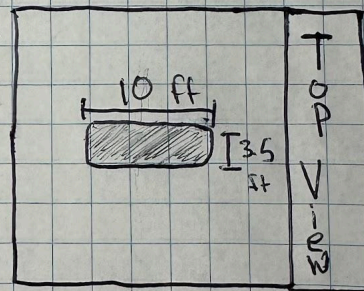
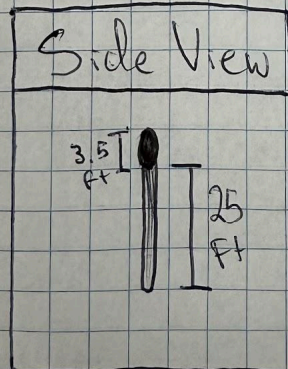
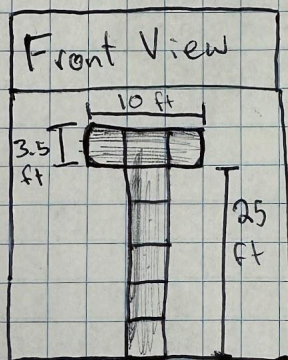
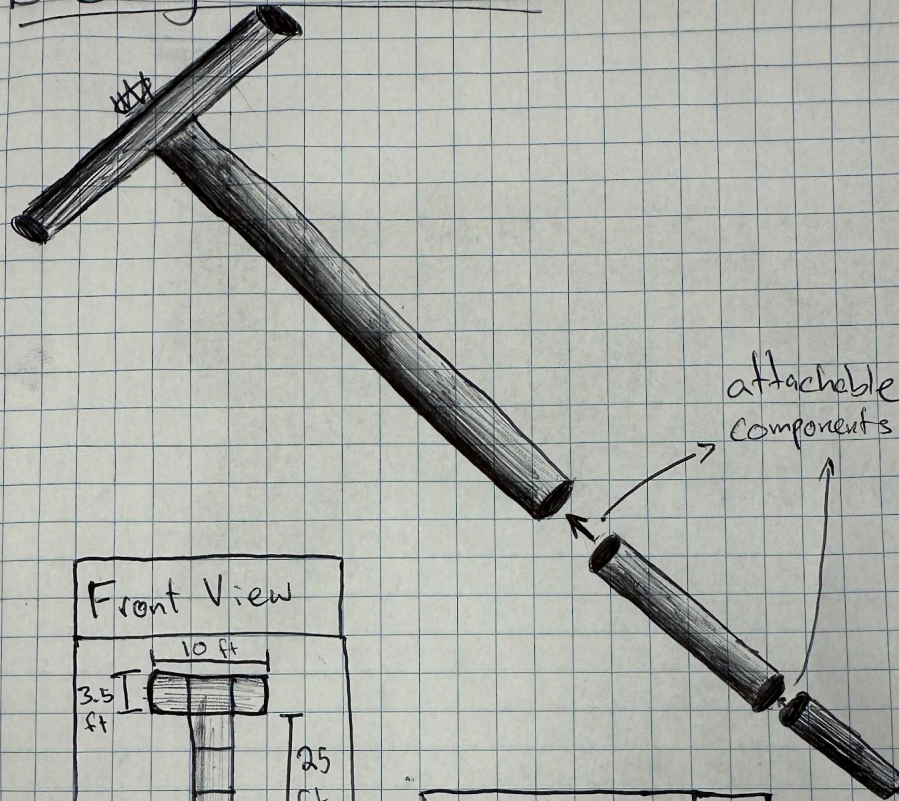
Design Idea #3



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Design Idea #3



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Manufacturing Methods:

- Step 1: Gather sturdy cardboard such as corrugated cardboard
- Step 2: Roll it up into two pieces; a 15 foot long piece, and a 10 foot piece. The 15 foot piece should have a **slightly** larger radius than the 10 foot piece
- Step 3: Slide the shorter piece into the longer piece.
- Step 4: Roll another 10 foot piece of cardboard and make a hole in the tube that fits the 15 foot piece's circumference

Step 5: Slide the whole thing into the second 10 ft piece and you have your stick!

Supply List and Costs:

Item	Cost each	# needed	Total
Cardboard	15 - 30 dollars	Aprox. 20 Ft	15-30 dollars
Tape	3 - 5 dollars	Aprox. 1 roll	3-5 dollars
Total Cost			18-35 dollars

Plan of Action (D3)

Plan of action Gantt chart:

	A	B	C	D
1	Weeks:	Assignments:	Design Elements	Status:
2	Feb. 5-9	Daily logs	Element D	Completed
3	Feb. 12-16	Daily logs	Element E/F	WIP
4	Feb. 19-23	Unknown	Unknown	Unknown
5	Feb. 26- Mar. 1	Unknown	Unknown	Unknown
6	Mar. 4-8	Unknown	Unknown	Unknown

Rubric

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Element D: Design Concept Generation, Analysis and Selection						
	5	4	3	2	1	0
1. Process for Generating & Comparing Possible Solutions. The Process for Generating Possible Solutions _____.	was <i>comprehensive, iterative, and consistently</i> defensible (viable design highly likely)	was <i>thorough, iterative, and</i> generally defensible (viable design likely)	was <i>adequate, generally</i> iterative and defensible (<i>viable design is possible</i>)	was <i>partial or overly general and only somewhat</i> iterative and/or defensible (raising issues with the design viability)	was <i>incomplete or minimally iterative and/or defensible</i> (raising issues with the design viability)	was not evident that there was an attempt
2. Design Solution Choice. The Design Solution chosen _____.	was <i>well-justified</i> and demonstrated attention to <i>all</i> design requirements	was <i>justified</i> and demonstrated attention to <i>most if not all</i> design requirements	was <i>explained with reference to at least some</i> design requirements	was not <i>sufficiently explained with reference to</i> design requirements	<i>lacked support related to</i> design requirements	was not attempted or evidence not provided
3. Plan of Action. The plan of action _____.	<i>has considerable merit and would easily</i> support repetition and testing for effectiveness by others	<i>would</i> support repetition and testing for effectiveness by others	<i>might not clearly or fully support repetition and testing for effectiveness by others</i>	<i>has insufficient detail to allow for testing for replication of results</i>	<i>is lacking to the point that the design solution cannot be tested</i>	cannot be executed or does not exist



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